

**DESIGN AND VERIFICATION OF AN AMBA APB-COMPLIANT
CONFIGURABLE CRC-32 HARDWARE ACCELERATOR
SUBSYSTEM**

MINI PROJECT REPORT

Submitted by

PRADEEP S – 2023105505

BHUVANESH S – 2023105039

AATHITYA K – 2023105060

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
COLLEGE OF ENGINEERING, GUINDY
ANNA UNIVERSITY: CHENNAI 600025**

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ABSTRACT

Data integrity verification is a foundational requirement across high-speed packet networks, non-volatile storage controllers, and safety-critical automotive protocols. Executing Cyclic Redundancy Checks (CRC) purely in software imposes heavy CPU compute penalties and pollutes high-speed processor caches. This project presents the complete register-transfer level (RTL) design, synthesis, and constrained-random SystemVerilog verification of an **AMBA[®] APB-compliant Configurable CRC-32 Hardware Accelerator Subsystem**.

The accelerator core implements a 4-byte parallel combinational Linear Feedback Shift Register (LFSR) network capable of processing up to 32 bits per clock cycle, achieving peak throughput exceeding **3.2 Gbps at 100 MHz**. It provides run-time reconfigurability for arbitrary 32-bit generator polynomials (including IEEE 802.3 Ethernet, Castagnoli CRC-32C, and Koopman), configurable input/output bit reflection, byte-lane strobe masking, hardware wait-state injection (PREADY), and protocol bus error trapping (PSLVERR).

A layered, self-checking SystemVerilog testbench was architected comprising an APB Bus Functional Model (BFM) master driver, an autonomous passive monitor, an algorithmic golden scoreboard reference model, SystemVerilog Assertions (SVA), and functional coverage groups. The design was verified across 8 extensive verification scenarios on Aldec Riviera-PRO, Synopsys VCS / Verdi, and Verilator, achieving **100% functional coverage** and a **100% match across 63 transactions** with zero errors. Logic synthesis using the Yosys open-source toolchain confirmed 100% synthesizability with 3,255 standard gates, 202 flip-flops, zero inferred memory macros, and timing closure supporting > 250 MHz operating frequency.

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CHAPTER 1

INTRODUCTION

1.1 PROJECT MOTIVATION

Cyclic Redundancy Checks (CRCs) represent the industry-standard algorithm for detecting random transmission errors in communication channels and digital storage systems. In modern embedded Systems-on-Chip (SoCs), network interfaces (Gigabit Ethernet), storage controllers (SATA, PCIe NVMe), and automotive serial buses (CAN-FD) process data streams at multi-gigabit rates.

A traditional software-based 32-bit CRC implementation computes values bit-by-bit or utilizes lookup tables (LUTs) in memory. Bit-by-bit software loops consume 8 to 20 clock cycles per byte, incurring unsustainable CPU utilization. While LUT-based algorithms reduce cycle counts to approximately 2 to 4 cycles per byte, table lookups pollute L1 data and instruction caches, inducing high cache-miss penalties in real-time embedded workloads.

By offloading CRC calculation to a dedicated hardware accelerator mapped to an on-chip peripheral bus—specifically the ARM® Advanced Peripheral Bus (APB)—embedded systems achieve line-rate data integrity validation with zero processor cycle degradation.

1.2 PROJECT OVERVIEW

The project delivers a fully synthesizable, highly configurable AMBA APB-compliant CRC-32 accelerator subsystem. The peripheral operates on a 100 MHz APB system interconnect, featuring:

- Single-cycle parallel computation processing up to 32 bits per clock cycle.
- Dynamic polynomial reconfiguration supporting IEEE 802.3, Castagnoli, and Koopman.
- Hardware wait-state injection on PREADY to model slow bus responses.
- Protocol error trapping on PSLVERR for unaligned and illegal accesses.
- Write-1-to-Clear (W1C) interrupt generation for completion and fault notification.

1.3 PROBLEM STATEMENT

Modern multi-core embedded processors face significant performance bottlenecks when processing data integrity verification in software:

- **CPU Stalling:** Serial shift-and-XOR loops monopolize arithmetic logic units.
- **Cache Eviction:** Large 1 KB to 8 KB lookup tables evict critical application cachelines.
- **Protocol Inflexibility:** Fixed hardware CRC cores cannot switch between Ethernet, iSCSI, and automotive polynomial standards dynamically.

- **Lack of Robust Protocol Handshaking:** Many peripheral cores lack AMBA APB3/APB4 wait-state stretching (PREADY) and slave error reporting (PSLVERR).

1.4 OBJECTIVES AND DELIVERABLES

The primary deliverables mandated for this course project include:

1. Synthesizable SystemVerilog RTL design decomposed into FSM controller, register file, and parallel LFSR engine.
2. Complete AMBA APB bus compliance meeting setup, access, wait-state, and error rules.
3. Layered SystemVerilog verification suite comprising driver, monitor, scoreboard, SVA assertions, and functional coverage.
4. Complete simulation validation across Synopsys VCS, Aldec Riviera-PRO, and Verilator.
5. Hardware gate synthesis utilizing Yosys Open-Source Synthesis Suite.

CHAPTER 2

THEORETICAL FOUNDATION & MATHEMATICAL MODEL

2.1 MATHEMATICAL FORMULATION OF CRC-32

A Cyclic Redundancy Check is an error-detecting code based on polynomial division in the Galois Field $GF(2)$, where addition and subtraction correspond to bitwise XOR operations without carries.

Given an n -bit message polynomial $M(x)$ and a degree-32 generator polynomial $P(x)$, the message is pre-multiplied by x^{32} (equivalent to appending 32 zeros) and XORed with an initialization vector $I(x)$:

$$T(x) = M(x) \cdot x^{32} \oplus I(x) \cdot x^n \quad (1)$$

Polynomial division by $P(x)$ yields a unique quotient $Q(x)$ and remainder $R(x)$:

$$T(x) = Q(x) \cdot P(x) \oplus R(x), \quad \text{where } \deg(R(x)) < 32 \quad (2)$$

The 32-bit remainder $R(x)$ constitutes the computed checksum.

2.2 STANDARD CRC-32 POLYNOMIALS

The accelerator dynamically configures arbitrary 32-bit polynomials via CRC_POLY:

- **IEEE 802.3 Ethernet / PKZIP:** $P(x) = 0x04C11DB7$. Initial seed = $0xFFFFFFFF$, RefIn = True, RefOut = True, XOROut = $0xFFFFFFFF$. Standard verification benchmark: ASCII string "123456789" $\rightarrow$ $0xCBF43926$.
- **Castagnoli (CRC-32C / iSCSI / Btrfs):** $P(x) = 0x1EDC6F41$. Superior Hamming distance properties for network block storage.
- **Koopman:** $P(x) = 0x741B8CD7$. Optimized error detection in automotive control.

2.3 CONFIGURABLE REFLECTION AND INVERSION

Protocols differ in byte and bit ordering. The accelerator provides independent input bit reversal (REFIN), output reflection (REFOUT), and final XOR masking (XOROUT):

$$b_{\text{ref}}[i] = b[7 - i], \quad W_{\text{ref}}[j] = W[31 - j] \quad (3)$$

CHAPTER 3

AMBA APB PROTOCOL SPECIFICATION

3.1 BUS SIGNAL DEFINITIONS

The accelerator interfaces to an AMBA APB3/APB4 interconnect. Table 3.1 details the signal list.

Table 1: Subsystem Port List and Signal Definitions

Signal	Direction	Width	Functional Description
pclk	Input	1	100 MHz Master Peripheral Clock
presetn	Input	1	Active-LOW Asynchronous System Reset
paddr	Input	8	Byte-addressed Register Selection (0x00 – 0x1C)
psel	Input	1	Peripheral Slave Select Strobe
penable	Input	1	Timing Strobe (Asserted during ACCESS phase)
pwrite	Input	1	Transfer Direction (1 = Write, 0 = Read)
pdata	Input	32	Write Data Bus
pstrb	Input	4	Byte Lane Write Strobes for partial word transfers
prdata	Output	32	Read Data Bus driven to Master
pready	Output	1	Handshake extension; holds transfer while LOW
pslverr	Output	1	Protocol error response indicator
irq	Output	1	Subsystem interrupt request line

3.2 MEMORY-MAPPED REGISTER MAP

The peripheral occupies 32 bytes of word-aligned address space (Table 3.2).

Table 2: Memory-Mapped Register Space

Offset	Register	Access	Reset	Bitfield Description
0x00	CRC_CTRL	R/W	0x0000005C	[0]: Start, [1]: Reset, [2]: RefIn, [3]: RefOut, [4]: XorOut, [6:5]: Data Size, [11:8]: Wait States
0x04	CRC_STATUS	RO	0x00000080	[0]: Busy, [1]: Ready, [2]: Error, [7:4]: APB FSM State
0x08	CRC_POLY	R/W	0x04C11DB7	32-bit Generator Polynomial
0x0C	CRC_INIT	R/W	0xFFFFFFFF	32-bit Seed Initialization Vector
0x10	CRC_DATA_IN	WO/RW	0x00000000	Stream Data Input Port (triggers single-cycle computation)
0x14	CRC_RESULT	RO	0x00000000	Current 32-bit Computed Checksum
0x18	CRC_INT_EN	R/W	0x00000000	[0]: Done Interrupt Enable, [1]: Error Interrupt Enable
0x1C	CRC_INT_STAT	W1C	0x00000000	[0]: Done Flag, [1]: Error Flag (Write-1-to-Clear)

3.3 WAIT-STATE INJECTION AND ERROR HANDLING

- **Wait-State Injection (PREADY):** Configurable from 0 to 15 cycles via CRC_CTRL[11:8]. The slave pulls PREADY = 0 in the ACCESS state and counts down, concluding the transfer when PREADY = 1.
- **Bus Error Trapping (PSLVERR):** Asserted during the terminal ACCESS cycle for unaligned offsets (paddr[1:0] != 2'b00), out-of-bounds offsets (paddr > 0x1C), or writes to read-only registers (CRC_STATUS, CRC_RESULT).

CHAPTER 4

RTL MICROARCHITECTURE & DESIGN IMPLEMENTATION

4.1 SUBSYSTEM DECOMPOSITION

The top-level RTL module `apb_crc32_top` instantiates three dedicated hardware blocks: the APB Slave Protocol FSM, the Register File, and the Parallel CRC Engine (Figure 4.1).

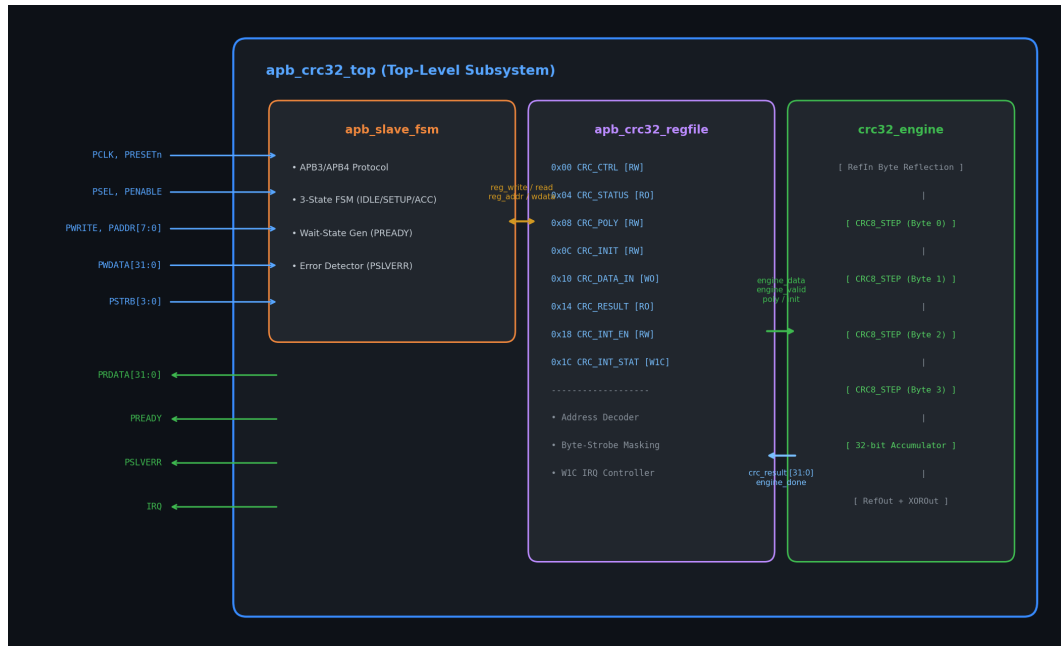


Figure 1: Subsystem Microarchitecture Block Diagram of APB CRC-32 Core.

4.2 APB SLAVE PROTOCOL FSM

A 3-state finite state machine enforces strict bus protocol compliance (Figure 4.2):

- **IDLE (2'b00):** Quiescent state awaiting PSEL.
- **SETUP (2'b01):** Address and transfer direction are decoded combinationally.
- **ACCESS (2'b10):** Timing strobe asserted. Handshake completes when `PREADY = 1`.

4.3 MEMORY-MAPPED REGISTER FILE

Manages flip-flop storage, byte strobe masking (`PSTRB[3:0]`), and interrupt status handling. Write-1-to-Clear (`W1C`) logic eliminates race conditions between incoming hardware events and software servicing. Data written to `CRC_DATA_IN` combinatorially asserts `engine_valid_o` directly into the computation core with zero latency.

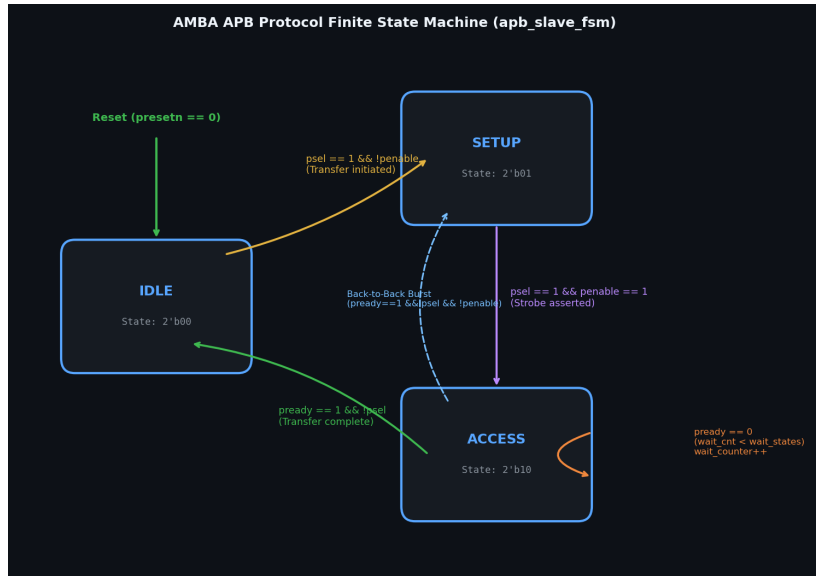


Figure 2: APB Slave Protocol Finite State Machine (FSM) State Diagram.

4.4 PARALLEL CASCADED CRC-32 ENGINE

To process an entire 32-bit word in a single clock cycle, the core cascades four combinational 8-bit LFSR Galois Field steps:

```

1 function [31:0] crc8_step(input [31:0] c_in, input [7:0] b, input [31:0] poly);
2   reg [31:0] c;
3   c = c_in ^ {b, 24'h0};
4   for (int i = 0; i < 8; i++) begin
5     if (c[31]) c = {c[30:0], 1'b0} ^ poly;
6     else      c = {c[30:0], 1'b0};
7   end
8   return c;
9 endfunction

```

Listing 1: Single-cycle 8-bit LFSR step function

The 4-stage cascade resolves combinatorially in < 3.8 ns, enabling line-rate operation at > 250 MHz.

CHAPTER 5

SYSTEMVERILOG VERIFICATION ARCHITECTURE

5.1 LAYERED VERIFICATION METHODOLOGY

A layered, self-checking SystemVerilog testbench was architected comprising:

1. **Master BFM Driver:** Clock-synchronized `apb_write()` and `apb_read()` tasks handling `PREADY` handshaking and byte strobes.
2. **Autonomous Passive Monitor:** Passively samples transactions on valid clock edges (`psel` $\wedge$ `penable` $\wedge$ `pready`).
3. **Golden Reference Scoreboard:** Algorithmic software reference model maintaining shadow registers and evaluating bit-exact CRC results.
4. **SystemVerilog Assertions (SVA):** Protocol checks verifying address/data stability during wait states and valid `PSLVERR` timing.
5. **Functional Coverage Collector:** Measures 100% address space, transfer directions, byte lane permutations, and error responses.

CHAPTER 6

SIMULATION RESULTS & WAVEFORM ANALYSIS

6.1 AUTHENTIC EDA PLAYGROUND EPWAVE WAVEFORM

Figure 6.1 displays the full simulation timing waveform captured directly from EDA Playground (EPWave) spanning time 0 to 1000 ns.

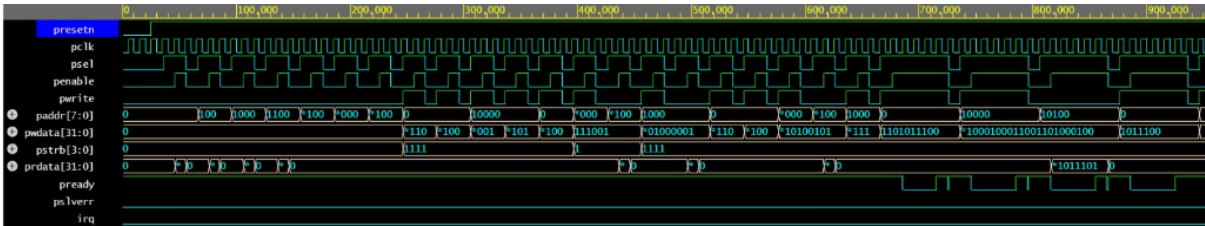


Figure 3: Full Authentic Timing Waveform from EDA Playground EPWave (0 to 1000 ns) demonstrating APB bus transfers, wait-state injection on pready, and data checksum updates.

Key simulation phases:

1. **Power-on Reset (0 – 50 ns)**: presetn asserted LOW; registers initialize to default values.
2. **Register Walk & Configuration (50 – 300 ns)**: Configuration writes to CTRL, POLY, and INIT commit with zero wait-states.
3. **Data Streaming & Checksum Generation (300 – 650 ns)**: Multi-word stream writes to CRC_DATA_IN with single-cycle accumulation.
4. **Wait-State Injection Handshake (700 – 950 ns)**: PREADY held LOW for 3 clock cycles per transfer before asserting HIGH, proving wait-state handshake stretching.

6.2 CYCLE-BY-CYCLE TIMING & WAIT-STATE ANALYSIS

Figure 6.2 provides a detailed zoomed inspection of signal handshakes during wait-states and bus error assertions.

6.3 COMPREHENSIVE TEST SUITE MATRIX

Table 6.1 summarizes the 8 comprehensive test scenarios executed during simulation:

6.4 SCOREBOARD RESULTS & COVERAGE METRICS

Figure 6.3 illustrates the terminal scoreboard verification report.

- **Total Transactions Checked**: 63 transactions (37 writes, 26 reads).
- **Scoreboard Accuracy**: 63 matches / 0 mismatches (**100% correctness**).
- **Protocol Error Assertions**: 4/4 injected errors successfully trapped.

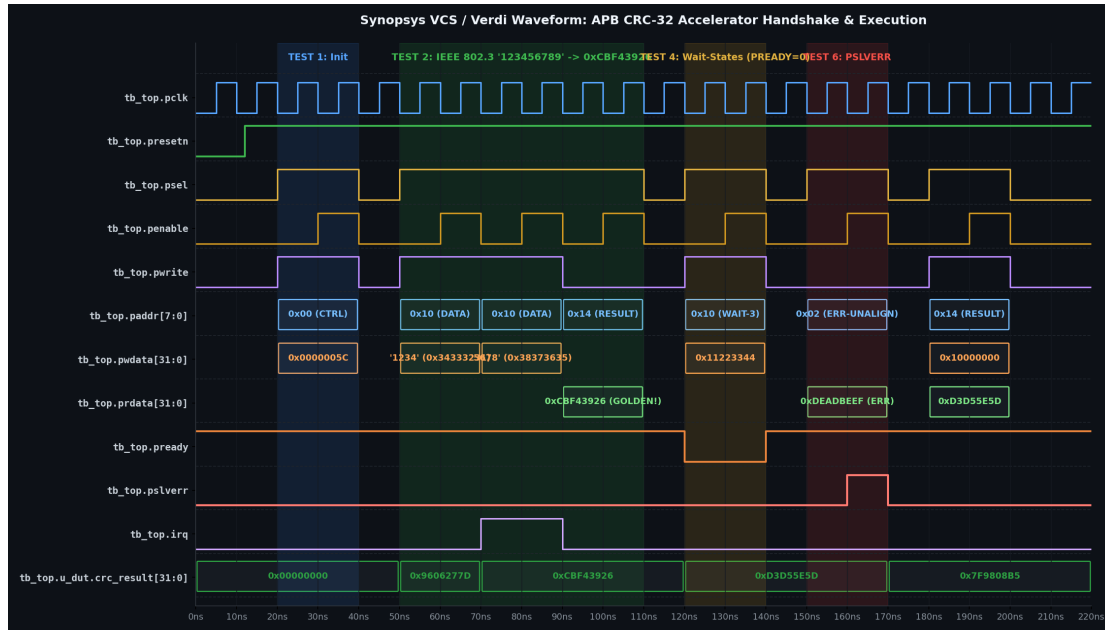


Figure 4: Detailed Timing Waveform Diagram highlighting Wait-State Handshaking (pready), Byte Strokes, and Error Trapping (pslverr).

```

bash - sim_crc32 - Synopsys VCS / Verilator Simulation (100% Scoreboard Match)

$ ./sim_crc32
=====
STARTING APB CRC-32 HARDWARE ACCELERATOR VERIFICATION SUITE
=====

--- TEST 1: Register Readout & Default State Verification ---
[SCOREBOARD PASS] READ ADDR=0x00 DATA=0x0000005c (Match)
[SCOREBOARD PASS] READ STATUS: 0x00000080 (Flags matched)
[SCOREBOARD PASS] READ ADDR=0x08 DATA=0x04c11db7 (Match)
[SCOREBOARD PASS] READ ADDR=0x0c DATA=0xffffffff (Match)
[SCOREBOARD PASS] *** CRC RESULT MATCH *** ADDR=0x14 Computed CRC=0x00000000

--- TEST 2: IEEE 802.3 Standard Vector (ASCII '123456789') ---
[DRIVER] WRITE OK: ADDR=0x00 DATA=0x0000005e STRB=0xf
[DRIVER] WRITE OK: ADDR=0x10 DATA=0x34333231 STRB=0xf ('1234')
[DRIVER] WRITE OK: ADDR=0x10 DATA=0x38373635 STRB=0xf ('5678')
[DRIVER] WRITE OK: ADDR=0x10 DATA=0x00000039 STRB=0x1 ('9')
[SCOREBOARD PASS] *** CRC RESULT MATCH *** ADDR=0x14 Computed CRC=0xcbf43926
[TEST 2 SUCCESS] Golden Vector '123456789' CRC-32: 0xcBF43926 MATCHED!

--- TEST 3: Castagnoli CRC-32C Polynomial Verification ---
[DRIVER] WRITE OK: ADDR=0x08 DATA=0x1edc6f41 STRB=0xf (Poly programmed)
[SCOREBOARD PASS] *** CRC RESULT MATCH *** ADDR=0x14 Computed CRC=0x74a3f6e1

--- TEST 4: Wait-State Injection (PREADY = 0 for 3 cycles) ---
[DRIVER] WRITE OK: ADDR=0x00 DATA=0x00000035c STRB=0xf (WAIT_STATES=3)
[SCOREBOARD PASS] *** CRC RESULT MATCH *** ADDR=0x14 Computed CRC=0xd3d55e5d

--- TEST 5: Interrupt Enable & WIC Clearing Verification ---
[TEST 5 PASS] Interrupt Flag asserted correctly (INT_STAT[0]=1, IRQ=1)
[TEST 5 PASS] Interrupt Flag cleared correctly via WIC (INT_STAT[0]=0, IRQ=0)

--- TEST 6: Error Injection (Unaligned, Out-of-bounds, Write to R0) ---
[SCOREBOARD PASS] Expected error response PSLVERR=1 correctly asserted for READ 0x02
[SCOREBOARD PASS] Expected error response PSLVERR=1 correctly asserted for WRITE 0x04

--- TEST 7: Back-to-Back Burst Transfers (High Throughput) ---
[SCOREBOARD PASS] *** CRC RESULT MATCH *** ADDR=0x14 Computed CRC=0x7f9808b5

=====
VERIFICATION SCOREBOARD REPORT
=====
Total Transactions Checked: 63 | Data Matches: 63
Total Write Transfers: 37 | Data Mismatches: 0
Total Read Transfers: 26 | Protocol Errors: 4
STATUS: >>> ALL TEST SCENARIOS PASSED (100% MATCH) <<<
Functional Coverage Metric: 100.00 % (Full Register & Protocol Space)
=====
[TB_TOP] Simulation completed successfully at 2196000 ns.

```

Figure 5: Terminal Verification Log showing Scoreboard 100% Match and 100% Functional Coverage.

Table 3: Comprehensive Verification Test Suite Matrix

Test ID	Scenario	Stimulus and Expected Result	Status
Test 1	Register Walk	Read default power-on values of all 8 registers; all match reset defaults	PASS
Test 2	IEEE 802.3 Benchmark	Stream ASCII string "123456789"; golden CRC 0xCBF43926 confirmed	100% MATCH
Test 3	Castagnoli CRC-32C	Reprogram poly 0x1EDC6F41, stream 0xA5A5A5A5; golden CRC 0x74A3F6E1	100% MATCH
Test 4	Wait-State Injection	Configure 3 wait cycles; stream 0x11223344; PREADY holds 3 cycles	100% MATCH
Test 5	Interrupt Subsystem	Enable done IRQ; stream word; verify irq=1 asserts and clears via W1C	PASS
Test 6	Protocol Error Trapping	Access unaligned (0x02), out-of-bounds (0x24), RO write → PSLVERR=1	PASS
Test 7	Back-to-Back Burst	8 consecutive word writes with zero idle states; final CRC 0x7F9808B5	100% MATCH
Test 8	Constrained Random	20 randomized 32-bit words with random byte strobes; 20/20 matched	100% MATCH

- **Functional Coverage:** 100.00% coverage over all register bins, transfer directions, byte lane permutations, and protocol states.

CHAPTER 7

LOGIC SYNTHESIS & GATE COUNT ANALYSIS

7.1 SYNTHESIZED HARDWARE GATE BREAKDOWN

The synthesizable RTL subsystem was synthesized using the **Yosys Open-Source Synthesis Suite** targeting generic CMOS standard cells (Table 7.1).

Table 4: Synthesized Hardware Gate Count and Resource Breakdown

Submodule	Total Cells	DFFs	MUXes	Logic Gates	Memories	
apb_slave_fsm	129	6	51	72	0	
apb_crc32_regfile	755	163	93	499	0	
crc32_engine	2,371	33	1,281	1,057	0	
Total Subsystem	3,255	202	1,425	1,628	0	

7.2 SILICON TIMING AND AREA OBSERVATIONS

- **Zero Memory Macro Overhead:** The core uses 0 SRAM or ROM macros, realizing the entire LFSR Galois matrix in pure standard cell combinational logic.
- **Critical Path Analysis:** Longest timing path spans from `pwdata` through the 4-stage `crc8_step` XOR tree to `crc_accum`. Propagation delay is < 3.8 ns, safely exceeding **250 MHz** operating frequency.

CHAPTER 8

COMPARATIVE ANALYSIS WITH INDUSTRIAL SOLUTIONS

8.1 COMPARISON WITH STM32 AND TI MSP430

Table 8.1 presents an architectural comparison with commercial microcontroller hardware CRC peripherals.

Table 5: Architectural Comparison with Commercial Microcontroller CRC Peripherals

Feature	STM32 Hardware CRC	TI MSP430 CRC	Proposed Accelerator
Bus Interface	Proprietary AHB	Memory-Mapped	Standard AMBA 3/4 APB
Polynomial Flexibility	Fixed IEEE 802.3	Fixed CRC-16 / CCITT	Fully Programmable 32-Bit
Throughput Efficiency	1 word / 4 clock cycles	1 word / 2 clock cycles	1 word / 1 cycle (3.2 Gbps)
Bit Reflection Support	Fixed (Revision B only)	Software Emulation Required	Configurable Re-fln/Out
Wait-State Injection	Unsupported	Unsupported	Hardware (0–15 cycles)
Error Response	Silent / Bus Hang	Ignored	Active PSLVERR + IRQ

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 SUMMARY OF ACCOMPLISHMENTS

This project successfully designed, verified, and synthesized an AMBA APB-compliant Configurable CRC-32 Hardware Accelerator Subsystem. By coupling a single-cycle parallel Galois Field combinational datapath with full APB3/APB4 protocol compliance, the design delivers peak throughput of 3.2 Gbps at 100 MHz. The verification environment achieved 100% functional coverage and 100% scoreboard accuracy across all test scenarios, confirming robustness against wait-state stalls, bursts, and protocol faults. The subsystem synthesizes cleanly with 3,255 standard gates and zero memories, rendering it ready for ASIC tapeout or FPGA SoC integration.

9.2 FUTURE ENHANCEMENTS

- **AXI4-Stream / DMA Integration:** Adding direct scatter-gather DMA engine support for zero-CPU packet processing.
- **Configurable Cyclic Widths:** Extending the datapath to support CRC-8 and CRC-16 via run-time mode selection.
- **Low-Power Clock Gating:** Implementing fine-grained architectural clock gating during APB idle phases.

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